Consumers, Producers, and Market Efficiency

Value an extra unit → build demand and supply → measure gains from trade → explain efficiency

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1. What does welfare economics ask?

A market decides *how many* units of a good are traded, *which buyers* receive them, and *which sellers* provide them. These choices matter because buyers value goods differently and sellers have different costs. The purpose of this chapter is to measure the gains from exchange and ask whether the market leaves any worthwhile trades unmade.

> **DEFINITION Welfare economics and total surplus**
>
> **Welfare economics** studies how the allocation of resources affects economic well-being. **Total surplus** measures the gains buyers and sellers obtain from voluntary trade: the value buyers place on what they receive, minus the opportunity cost of providing it.

Consider a repair service that a customer values at \$50 and a technician can provide at a full opportunity cost of \$30. If they trade, society gains \$20: the service is worth \$20 more to the buyer than the resources given up to produce it. If they do not trade, that possible gain is lost. The price determines *how they divide* the \$20, not whether the trade creates it.

> **KEY RESULT The central question**
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> An additional unit creates a gain from trade when the buyer's **willingness to pay exceeds the seller's opportunity cost**. The market is efficient if it realizes all such gains, without producing units whose costs exceed their values. We will show why this happens in a competitive market under specified assumptions.

2. Diminishing marginal utility and maximum willingness to pay

To understand demand, first think about the value of *one more* unit. One visit to a quiet study room may be especially useful before an examination. A second visit may still be helpful, but less urgent. Additional visits can provide progressively smaller benefits. This is the intuition behind diminishing marginal utility.

> **DEFINITION Total utility, marginal utility, and diminishing marginal utility**
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> **Total utility** is the overall satisfaction or benefit from consuming a quantity of a good. **Marginal utility (MU)** is the **additional** satisfaction from consuming one more unit.
> The **principle of diminishing marginal utility** says that, as consumption of a good rises, the satisfaction from an additional unit *typically* falls, other things equal. It does not say that total satisfaction must fall.

Worked example: Additional study-room visits

Suppose the following table describes the benefits from monthly visits, measured in illustrative *utility units*. The numbers are hypothetical; utility units are not dollars.

Visit	Marginal utility of that visit	Total utility after the visit
1st	18	18
2nd	12	30
3rd	7	37
4th	3	40

The second visit adds 12 utility units, raising total utility from 18 to 30. The fourth adds only 3, raising total utility from 37 to 40. **Marginal utility falls while total utility still rises.** Total utility would begin to fall only if an additional visit had *negative* marginal utility.

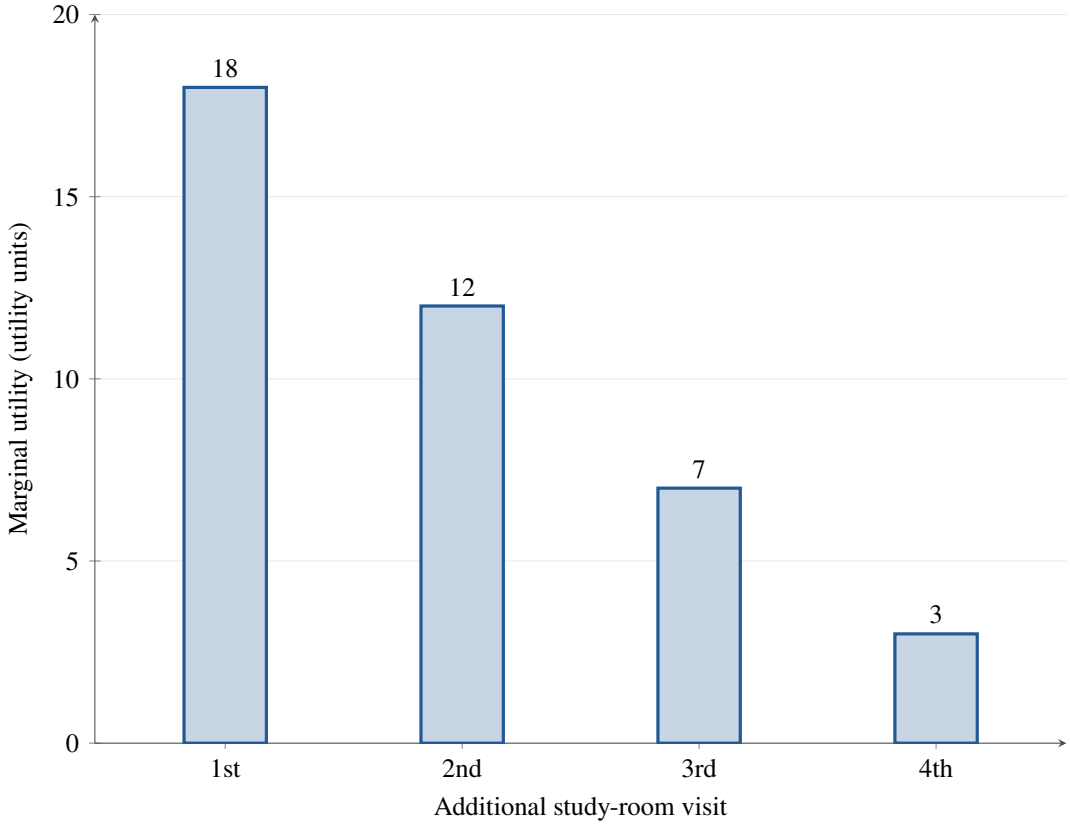


Figure 1. Diminishing marginal utility. Each successive visit adds positive satisfaction, but the *additional* benefit becomes smaller.

Read the graph. Each bar is the satisfaction from that one visit, not the total from all visits. The bars fall from 18 to 3. The table shows that total utility nevertheless rises from 18 to 40. **Lower marginal utility is not the same as lower total utility.**

DEFINITION Maximum willingness to pay (WTP)

A buyer's **maximum willingness to pay** is the highest amount of money the buyer would pay for a unit rather than go without it. For several units, each additional unit may have a different maximum WTP. A buyer purchases another unit when its WTP is at least the price, assuming the buyer can afford the purchase.

Diminishing marginal utility helps explain why a buyer often has declining WTP for successive units. Satisfaction is measured in utility, while WTP is measured in dollars; they are **not the same unit**. For introductory demand analysis, we use WTP as the buyer's *money measure* of the benefit of another unit. The precise conversion depends on preferences and the value of money to that buyer.

3. From willingness to pay to individual demand

Suppose a student can buy visits to a climbing gym one at a time during a month. The student values the first visit at $32, the second at $24, the third at $16, and the fourth at $8. These are maximum amounts the student would pay *for each additional visit*; they are not the total value of all visits together.

Price per visit	$35	$28	$20	$12	$6
Visits demanded	0	1	2	3	4

At $20, the first two visits are worth at least the price, but the third is worth only $16. The student buys **two**. At $12, three visits are worthwhile. The *law of demand* follows: as the price falls, holding other influences fixed, the quantity demanded typically rises.

DEFINITION Individual demand and the marginal buyer

An **individual demand schedule** shows how many units one buyer is willing and able to buy at different prices. The **demand curve** graphs that relationship. At a given quantity, its height shows the **WTP for the last unit purchased**: the marginal unit. With only a few units, demand looks like a staircase.

KEY RESULT Why demand typically slopes downward

As a person consumes more units, the added benefit normally diminishes. The buyer is therefore willing to buy later units only at lower prices. Changes in *price* move along a given demand curve; changes in income, tastes, or other demand determinants can *shift* the curve.

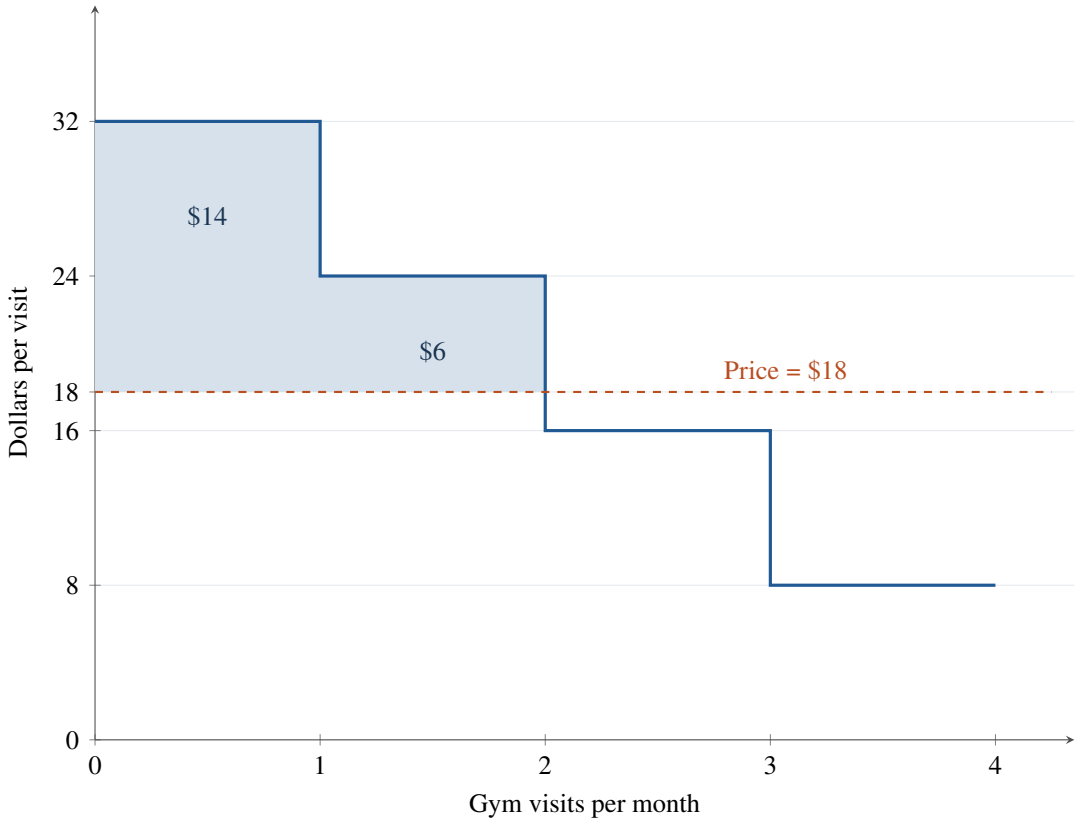


Figure 2. Individual demand from maximum willingness to pay. At a price of $18, two visits are purchased; the shaded rectangles are the gains to the buyer.

Read the graph. The blue steps are the WTP for the first, second, third, and fourth visits. The orange line is the $18 price. Only the first two steps are above that line. The difference between WTP and price on those two units measures the buyer's **consumer surplus**, explained below.

4. From individual demand to market demand

A market contains many buyers. At each price, add the *quantities* each buyer wants, not the prices or their dollar valuations. This is called **horizontal addition** of individual demand curves.

Suppose a second buyer's WTP for three visits is $27, $17, and $7. Both buyers face the same admission price. The table shows how to build market demand from their individual choices.

Price	Visits: buyer A	Visits: buyer B	Market visits
$30	1	0	**1**
$22	2	1	**3**
$18	2	1	**3**
$15	3	2	**5**
$10	3	2	**5**
$5	4	3	**7**

At $18, buyer A purchases two visits and buyer B purchases one. Market quantity demanded is 2 + 1 = 3 **visits**. At $15, the quantities are three and two, making five in total.

> **KEY RESULT The market-demand rule**
>
> **At each price, market quantity demanded = sum of all individual quantities demanded.** A market-demand curve slopes downward in the usual case because a lower price brings in more buyers or induces existing buyers to buy more units. The height of market demand at a given quantity represents the WTP of the *marginal buyer or unit*.

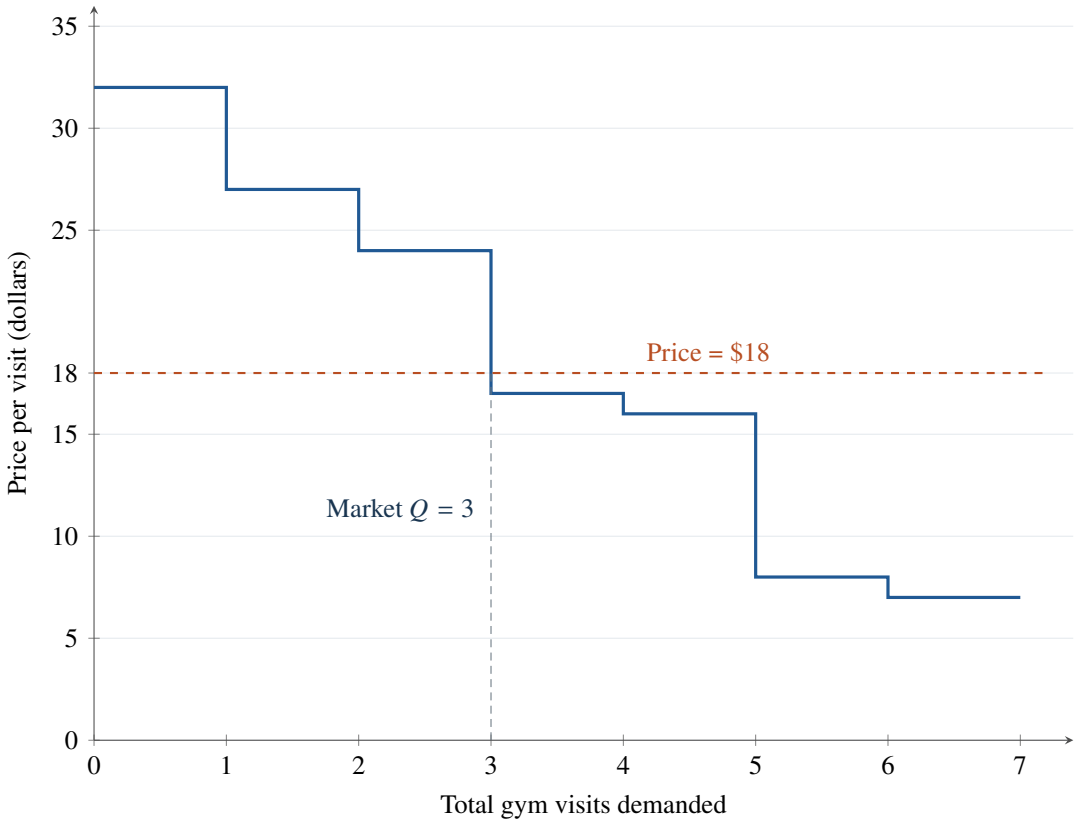


Figure 3. Market demand. Order every potential purchase by its willingness to pay; at $18, three of the seven possible visits are valued highly enough to be bought.

Read the graph. There are seven steps because the two buyers together might buy up to seven visits. At $18, the price lies below the WTP of three units but above the WTP of the fourth. **Three visits are demanded.** With many buyers, the steps tend to be small and the market-demand curve looks smooth.

5. Consumer surplus: measuring buyers' gains

Demand tells us which purchases buyers choose. Consumer surplus measures the gains they get from those purchases at the market price.

DEFINITION Consumer surplus (CS)

Consumer surplus is **maximum willingness to pay minus the price actually paid**. For one unit,

$$\boxed{\text{Consumer surplus} = \text{WTP} - \text{price}.}$$

Market consumer surplus is the sum of this difference over all units bought. On a price–quantity graph, it is the **area under demand and above the price line**, up to the quantity purchased.

Return to the gym visitor in Section 3. At \$18 per visit, WTP is \$32 for the first visit and \$24 for the second. The buyer gets \$14 on the first and \$6 on the second, for **\$20 total CS**. This is not a cash payment from the gym. It is the buyer's measured benefit from paying less than the maximum the visits were worth.

Worked example: Consumer surplus with a smooth demand curve

To learn the geometry, consider a *separate hypothetical market* with a straight demand curve. Suppose the highest WTP is \$54 and demand reaches the price \$30 at a quantity of 12. The triangle between demand and price has a **base of 12 units** and a **height of** \$54 − \$30 = \$24.

$$\text{Consumer surplus} = \tfrac{1}{2} \times 12 \times \$24 = \boxed{\$144}.$$

Why a triangle? The first units are worth more than the price. For each later unit, WTP falls until the last buyer is just willing to pay \$30. Adding up the differences between WTP and price gives the whole shaded area.

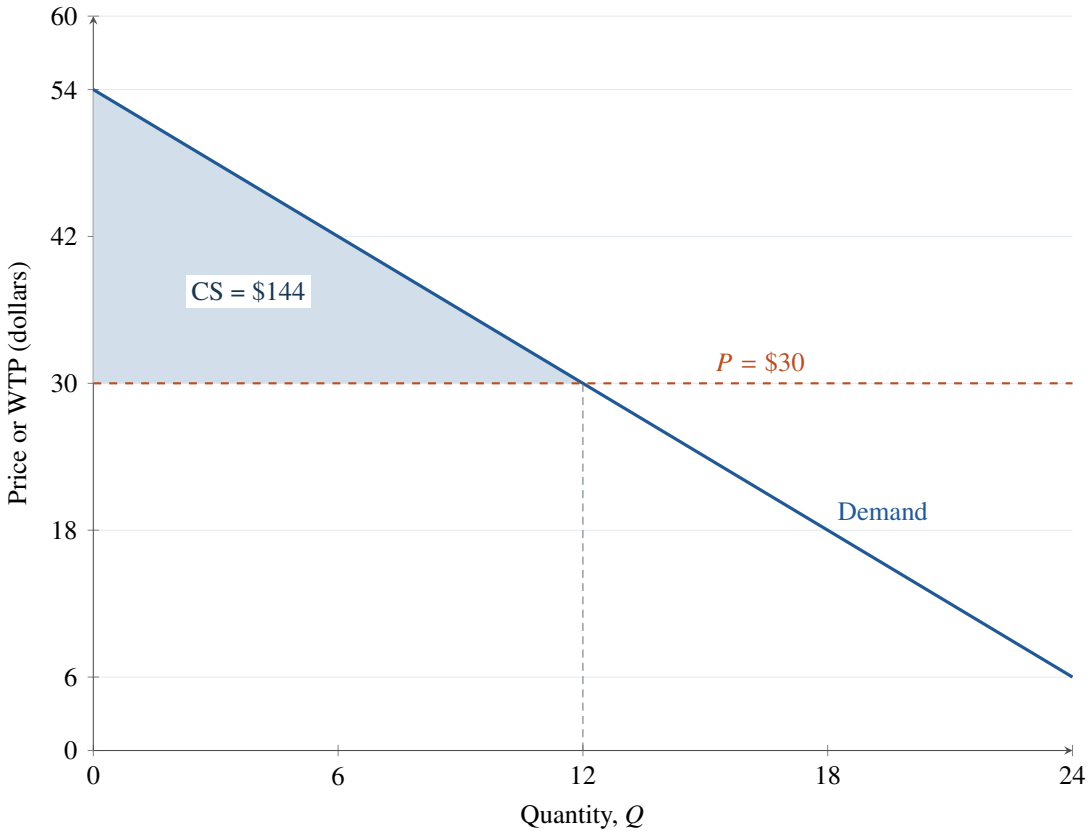


Figure 4. Consumer surplus in a smooth market. The shaded triangle measures how much buyers' willingness to pay exceeds the price on the 12 units purchased.

Read the graph. Demand starts at \$54 and meets the price line at \$30 when $Q = 12$. The triangle is *above* price and *below* demand. Using the full \$54 as the triangle's height would be wrong: the height is \$24, after subtracting the price.

> **KEY RESULT What a price change does to consumer surplus**
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> When the price falls, **existing buyers gain** because they pay less for units they already bought, and **additional buyers may enter** because some purchases not worth making at the old price become worthwhile. Both can raise CS. When the price rises, the two effects work in reverse.

Worked example: Separating the two gains from a price cut

Using the smooth demand curve above, suppose the price falls from \$36 to \$30. Quantity demanded rises from 9 to 12. At \$36, $CS = \frac{1}{2}(9)(\$54 - \$36) = \$81$. At \$30, CS is \$144. The **gain is \$63**: the original 9 units become \$6 cheaper, giving \$54, while the 3 additional units generate \$9 in new CS (a triangle with base 3 and height \$6). Thus \$54 + \$9 = \$63. These prices illustrate a movement along demand; a change in market equilibrium would also depend on supply.

6. Producer surplus: measuring sellers' gains

Buyers have maximum WTP. Sellers have a corresponding *minimum willingness to sell*. A seller must cover the opportunity cost of producing a unit, including the value of time and any other resources given up.

> **DEFINITION Cost, willingness to sell, and producer surplus**
>
> **Willingness to sell (WTS)** is the lowest price a seller would accept for a unit. In this chapter's simple model, it equals the seller's **opportunity cost**. **Producer surplus (PS)** is
>
> $$\boxed{\text{Producer surplus} = \text{price received} - \text{opportunity cost.}}$$
>
> Market producer surplus is the sum across units sold: on a graph, it is the **area above supply and below the price line**, up to quantity sold.

Worked example: Independent office-chair repair technicians

Three technicians can each complete one repair appointment. Their costs, including the value of their time, are \$9, \$15, and \$23. At a price of \$19, the first two offer appointments but the third does not. Their producer surpluses are \$19 – \$9 = \$10 and \$19 – \$15 = \$4: **total PS is \$14**. The third technician would lose \$4 and therefore declines to sell.

> **KEY RESULT Why supply curves typically slope upward**
>
> To expand output, a market often draws on progressively higher-cost units or sellers. Therefore, a higher price makes additional production worthwhile. At each quantity, the height of supply represents the **opportunity cost of the marginal unit**. Cost here means *economic cost*, not just cash expenses.

Worked example: Producer surplus with a smooth supply curve

In a separate hypothetical market, suppose supply begins at a marginal cost of \$6 and rises to \$30 at a quantity of 12. At price \$30, producers sell 12 units. The area *above supply* and *below price* is a triangle with a base of 12 and a height of \$30 − \$6 = \$24:

$$\text{Producer surplus} = \tfrac{1}{2} \times 12 \times \$24 = \boxed{\$144}.$$

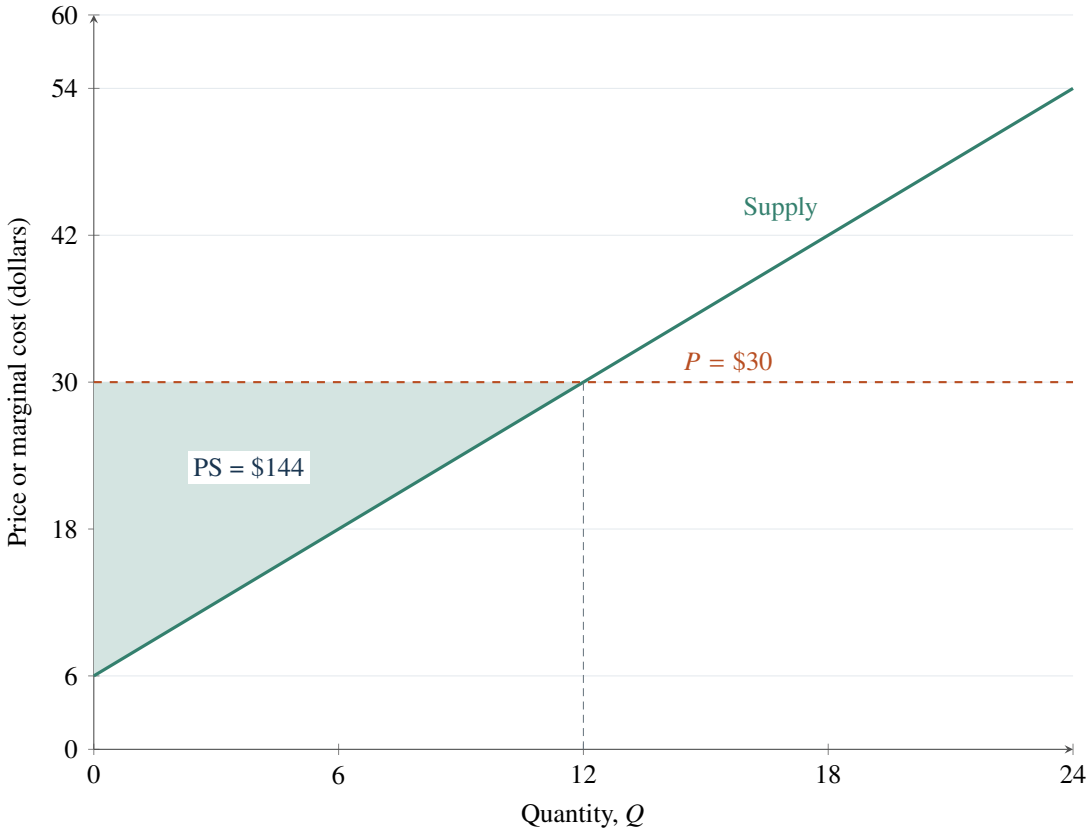


Figure 5. Producer surplus in a smooth market. The shaded triangle measures the payment producers receive above their opportunity costs on 12 units.

Read the graph. The supply curve is the marginal cost of providing successive units. The first units cost much less than the \$30 price and generate large gains to sellers. The last unit's cost reaches the price. A higher price normally raises PS through higher receipts on existing sales and additional production; a lower price does the reverse.

Worked example: Separating the gains from a higher selling price

On the same supply curve, suppose the price rises from \$30 to \$36. At \$36, sellers supply 15 units, and producer surplus becomes $\tfrac{1}{2}(15)(\$36 - \$6) = \$225$, up from \$144. The **gain is \$81**: sellers of the original 12 units receive \$6 more on each (a \$72 gain), while the 3 additional units produce \$9 in new PS (a triangle with base 3 and height \$6). Thus \$72 + \$9 = \$81. As with the consumer example, this compares different prices *along a given supply curve*; it does not describe an entire equilibrium adjustment.

7. Total surplus and efficiency of competitive markets

We can now put buyers and sellers together. A buyer's surplus is the value of a purchase less the price paid; a seller's surplus is the price received less the cost. Add them: the price cancels, because one person's payment is another person's receipt.

KEY RESULT The total-surplus rule

$$\boxed{\text{Total surplus} = \text{CS} + \text{PS} = \text{value to buyers} - \text{cost to sellers.}}$$

An **efficient allocation** maximizes total surplus. **A trade adds to total surplus whenever WTP exceeds opportunity cost.** If cost exceeds WTP, producing that unit destroys surplus.

Worked example: Which transactions should take place?

Three customers each want one data-recovery appointment. Their maximum WTP values are \$45, \$35, and \$25. Three independent technicians can each supply one appointment at opportunity costs of \$12, \$28, and \$39. The most valuable buyers are served first, and the lowest-cost technicians work first.

Potential trade	Buyer's WTP	Seller's cost	Gain from trade	Decision
1st	\$45	\$12	**\$33**	Trade
2nd	\$35	\$28	**\$7**	Trade
3rd	\$25	\$39	**−\$14**	Do not trade

Two appointments are efficient: \$33 + \$7 = \$40 in total gains. The third would cost \$39 but be worth only \$25; forcing it would reduce total surplus by \$14. At a common market price of \$30, the two highest-value customers buy, the two lowest-cost technicians sell, and **two trades occur**. Buyers' CS is \$15 + \$5 = \$20 and sellers' PS is \$18 + \$2 = \$20. Total surplus is \$40. Different prices can divide the gain differently without changing the \$40 total, as long as the same efficient trades occur.

KEY RESULT Three properties of the competitive equilibrium

Under the usual assumptions of many price-taking buyers and sellers, no externalities, and enforceable property rights:

1. The goods go to the buyers with the highest willingness to pay.

2. Production is supplied by sellers with the lowest opportunity costs.

3. The market trades the quantity that maximizes the combined gains from exchange, **CS + PS**. These are efficiency claims, not claims about equality.

8. See efficiency on the supply-and-demand graph

Use one more *hypothetical smooth market* to put the argument together. Suppose the demand curve is $P = 54 - 2Q$ and the supply curve is $P = 6 + 2Q$, with prices in dollars. You do not need calculus. At $Q = 12$, buyers' WTP and sellers' marginal cost are both \$30. Thus the market equilibrium is **12 units at \$30**.

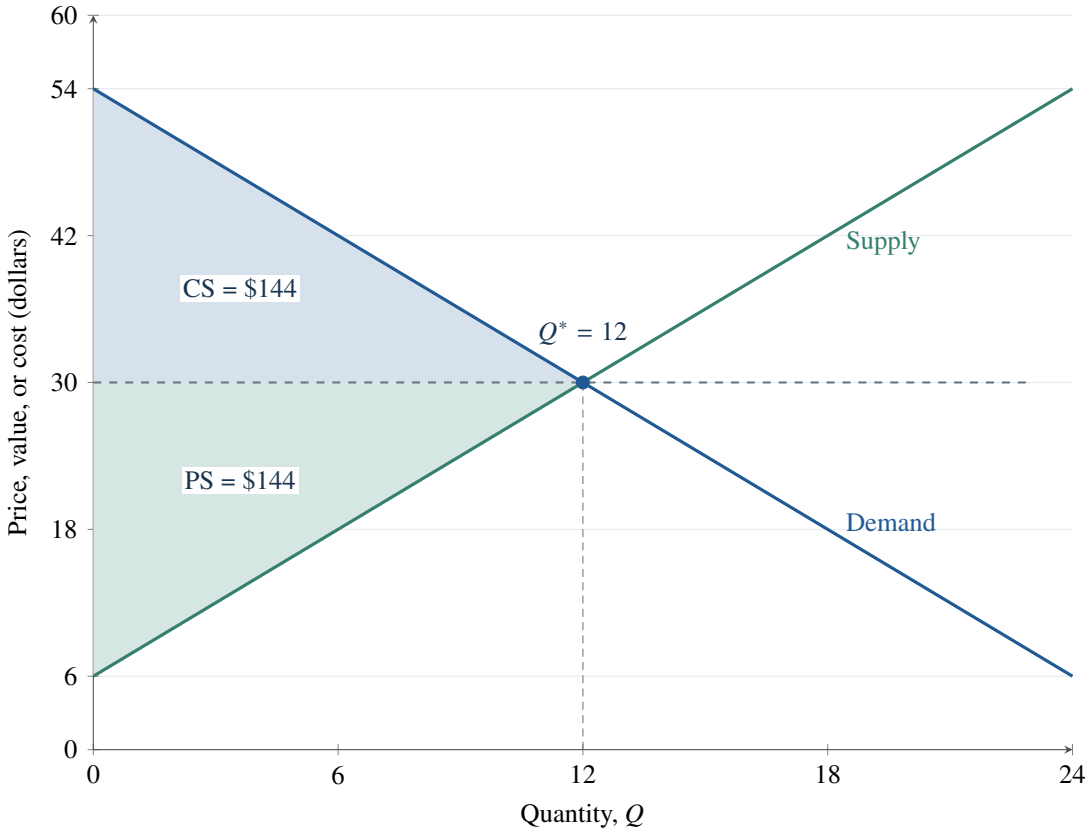


Figure 6. Competitive-market efficiency. Demand measures marginal willingness to pay; supply measures marginal cost. They meet at 12 units, where all available gains from trade have been realized.

Read the graph. To the *left* of 12, the blue demand curve is above the green supply curve: another unit is worth more than it costs, so more trade raises total surplus. To the *right* of 12, cost exceeds value: producing more would reduce total surplus. At 12, the two are equal. The blue area is consumer surplus and the green area is producer surplus. Together, \$144 + \$144 = \$288.

Worked example: Why both too little and too much trade are inefficient

At $Q = 9$, the demand price is \$36 and marginal cost is \$24. Another unit creates roughly \$12 more value than cost at that point, so stopping there leaves useful trades unmade. At $Q = 15$, the demand price is \$24 and marginal cost is \$36. Additional production around that point costs roughly \$12 more than buyers value it, so producing that much wastes resources. Neither deviation improves on $Q = 12$.

9. Efficiency, equality, and the limits of the result

KEY RESULT The competitive-market efficiency result

With competitive buyers and sellers and no relevant market failures, equilibrium **maximizes total surplus**. It does not maximize the well-being of each person separately, and it does not imply that every person receives an equal share. The result is conditional on the model's assumptions.

DEFINITION Efficiency versus equality

Efficiency asks whether all possible gains from trade have been realized and total surplus is as large as it can be. **Equality** asks how evenly economic well-being is distributed across people. A market can be efficient without producing an equal distribution; measures intended to change distribution may also affect incentives and total surplus.

The idea often associated with Adam Smith's *invisible hand* is that buyers and sellers pursuing their own interests can coordinate through prices and generate valuable trades without a central planner choosing each purchase or sale. Our demand-and-supply analysis explains *why* this can work: buyers compare price with value, and sellers compare price with cost.

But the proof has limits. With **market power**, price need not equal marginal cost and some mutually beneficial trades may not occur. With **externalities**, the private demand or supply curves can omit benefits or costs borne by others. In those cases a competitive-looking market outcome need not maximize *social* surplus. Other institutions, information problems, or restrictions on exchange can also change the result.

Why a restriction can matter. Suppose access to a scarce service is rationed by a waiting list and cannot be resold. The people first in line need not be those who value the service most. If price is held below the market-clearing level, the number seeking service can exceed the number available. But efficiency alone does not determine how society should weigh fairness, legal rights, or other effects. Welfare economics tells us what the simple market model predicts; it does not settle every policy question.

KEY RESULT What to take from Chapter 7

Demand: Diminishing marginal utility helps explain declining maximum WTP for additional units. At each price, add individual *quantities* to obtain market demand.
Consumer surplus: WTP minus price; area *below demand* and *above price*.
Producer surplus: Price minus opportunity cost; area *above supply* and *below price*.
Market efficiency: Total surplus is CS + PS, or total value minus total opportunity cost. Competitive equilibrium maximizes it *under the model's assumptions*. Efficiency and equality answer different questions.